Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2013-0451	Issued	2013-12-09

UT Thickness Re-measurement - Reflux Lines 3-RF-101 and 2-RF-102

Scope and Responsibilities

This report documents the ultrasonic thickness re-measurement of reflux lines 3-RF-101 and 2-RF-102 within the ATU-100. The scope was limited to correcting data discrepancies identified in the previous survey of the regenerator overhead and reflux systems. Kevin Doucette, acting as the lead UT technician, was responsible for the execution of the field measurements and the verification of equipment calibration against the required material velocity.

The inspection was performed under QSCP-NDT-001 Rev 2, which incorporates updated requirements for insulation removal and CUI program compliance. All measurements were conducted in accordance with ASTM E797 to ensure high-resolution data acquisition. The primary objective was to replace the rejected data from the previous 3-year survey cycle with accurate readings calibrated for the specific alloy properties of the piping.

Background and Nonconformance

This document responds to QSCP-NCR-2013-019, which was issued following the discovery that previous readings for the 316SS reflux lines were not reproducible. The nonconformance report noted that carbon-steel velocity settings were incorrectly applied during the data collection for HIS-2013-0402. Consequently, the Hamilton Inspection Department rejected the previous data set, necessitating a full re-measurement of the affected circuits.

The lines in question, 3-RF-101 and 2-RF-102, are constructed from A312 TP316 stainless steel. Because the ultrasonic velocity of stainless steel differs significantly from carbon steel, the initial readings recorded in HIS-2013-0402 were technically invalid for asset integrity calculations. This re-measurement ensures that the retirement thickness and remaining life assessments are based on accurate physical measurements.

Technical Data and Line Facts

Line 3-RF-101 is an 88.9 mm OD pipe with a nominal thickness of 5.49 mm and a design t-min of 2.41 mm. The design limits for this section are 517 kPa(g) and 104 °C. During this mobilization, Kevin Doucette verified the instrument velocity setting for 316SS prior to taking readings at all established CML locations to ensure compliance with the site rules for Class 2 reflux service.

Line 2-RF-102 has an OD of 60.3 mm and a nominal thickness of 3.91 mm, with a design t-min of 2.16 mm. It shares the same design pressure of 517 kPa(g) and design temperature of 104 °C as the larger reflux line. All re-measured values were compared against these t-min values to determine if any immediate nonconformance existed, as any reading below t-min requires an NCR per site integrity protocols.

Integrity Assessment and Scheduling

The corrected data indicates that both 3-RF-101 and 2-RF-102 remain above their respective retirement thicknesses. Per the site rules for Class 2 amine and reflux service, the maximum thickness inspection interval is 10 years; however, the regenerator overhead circuit is maintained on a 3-year thickness interval. The remaining life was calculated using the short-term corrosion rate derived from this re-measurement and the valid historical data points.

In accordance with QSCP-NDT-001 Rev 2, these lines are also subject to the CUI program requirements. Since these are insulated lines, they require profile radiography or insulation windows every 5 years. The next thickness reading for these ATU-100 components will be scheduled at no more than half the remaining life, ensuring the integrity of the 316SS piping remains within the 517 kPa(g) design envelope.

Thickness readings, measured 2013-12-09

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2013-09-09	5.39	5.37	0.006	0.014	>50	2023-12-09	
3-RF-101-02	5.49	2.41	2013-09-09	5.34	5.40	0.003	-0.004	>50	2023-12-09	
3-RF-101-03	5.49	2.41	2013-09-09	5.45	5.37	0.005	0.010	>50	2023-12-09	
3-RF-101-04	5.49	2.41	2013-09-09	5.33	5.41	0.003	-0.010	>50	2023-12-09	
3-RF-101-05	5.49	2.41	2013-09-09	5.26	5.32	0.007	0.010	>50	2023-12-09	
3-RF-101-06	5.49	2.41	2013-09-09	5.25	5.17	0.020	0.030	>50	2023-12-09	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-07	5.49	2.41	2013-09-09	5.43	5.48	-0.003	-0.017	>50	2023-12-09	
3-RF-101-09	5.49	2.41	2013-09-09	5.12	5.26	-0.562	-0.562	>50	2023-12-09	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2013-09-09	3.71	3.60	0.018	0.032	44.9	2023-12-09	
2-RF-102-02	3.91	2.16	2013-09-09	3.65	3.70	0.010	0.023	>50	2023-12-09	
2-RF-102-03	3.91	2.16	2013-09-09	3.76	3.79	0.008	0.001	>50	2023-12-09	
2-RF-102-04	3.91	2.16	2013-09-09	3.80	3.82	0.003	-0.001	>50	2023-12-09	
2-RF-102-05	3.91	2.16	2013-09-09	3.69	3.67	0.080	0.080	18.8	2023-05-05	
2-RF-102-06	3.91	2.16	2013-09-09	3.71	3.67	0.161	0.161	9.4	2018-08-22	
2-RF-102-07	3.91	2.16	2013-09-09	3.82	3.72	0.014	0.017	>50	2023-12-09	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2013-12-09	_____